

Relative Exponential Periods on Polynomial Flags: Polynomial Boundary Systems and an Elliptic Comparison Theorem

Christopher D. Long

Antoine-Auguste Le Blanc*

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galizur@gmail.com

Abstract

We construct rational differential systems for compact polynomial-flag exponential integrals using polynomial representatives in twisted de Rham cohomology. Every forcing term is an actual compact polynomial-line face integral, with its normalization retained by a literal Stokes identity. An actual-solution criterion upgrades a nonzero quotient by the linear boundary space to algebraic independence over the full boundary field, under special-linear Galois and projective-disjointness hypotheses. For the triangle with vertices $(-1/2, -1)$, $(1/2, 0)$, $(-1/2, 1)$ and phase $y^2 - x^3 + 3x$, these hypotheses are verified. The two moments with amplitudes 1 and x are algebraically independent over the field of the three fixed faces. Their values are algebraically independent over the associated face-value field at every nonzero algebraic parameter. Every polynomial amplitude has a unique two-coordinate Stokes remainder.

We retain the single-phase denominator-free comparison on the observable Lefschetz locus, face-lattice and Hodge statements with their correct variance and coefficient fields, and an abstract multicolumn orbit calculation. For general local comparison, Beukers lifting identifies the boundary evaluation prime directly, without a global normal-form claim. Numerical injectivity is formulated on the specialized fiber, not on the unspecialized function algebra. General multiblock formal completeness requires separately verified actual linear and determinant-line realizations; it is not inferred from punctual support or observability.

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1 Introduction and theorem structure

Let $k = \overline{\mathbb{Q}}$ and $k_{\mathbb{R}} = k \cap \mathbb{R}$. A polynomial two-chain is a finite k -linear combination of polynomial pushforwards of $[0, 1]^2$, with orientation and multiplicity. A flag is the special case with a triangular parametrization. For a polynomial phase q and a polynomial two-form ω , put

$$I_D(\omega)(z) = \int_D e^{zq} \omega.$$

Kontsevich–Zagier relations are generated by algebraic changes of variables, linearity and Stokes; in the exponential setting one uses $d_z = d + z dq \wedge$ [9, 4]. The compact polynomial-line comparisons are developed in the companion Papers II and III [10, 11].

A boundary field here always refers to the finitely many *fixed faces* of the specified chain. It is not the field of all compact line periods. A finite closed vector of actual face functions generates it over $k(z)$. The construction in Theorem 2.1 makes all inhomogeneous forcing terms elements of this field by polynomial identities, rather than by assigning arithmetic normalizations to abstract composition factors.

The main concrete case is

$$q = y^2 - x^3 + 3x, \quad D = \text{conv}\{(-1/2, -1), (1/2, 0), (-1/2, 1)\}, \quad M_j = I_D(x^j dx \wedge dy) \quad (j = 0, 1). \quad (1.1)$$

Let $K_{\partial,0}$ be its actual face-function field over $k(z)$ and $K_{\partial,\xi}$ its face-value field at $\xi \in k^\times$.

Theorem 1.1 (Elliptic functional and numerical comparison). *For (1.1), M_0, M_1 are algebraically independent over $\mathbb{C}K_{\partial,0}$. For every $\xi \in k^\times$, the values $M_0(\xi), M_1(\xi)$ are algebraically independent over $K_{\partial,\xi}$. Every $r \in k[x, y]$ has a unique two-coordinate remainder and a polynomial Stokes certificate*

$$r = a_r(z) + b_r(z)x + L_x P_r + L_y Q_r,$$

where $a_r, b_r \in k[z, z^{-1}]$, $P_r, Q_r \in k[z, z^{-1}, x, y]$, $L_x = \partial_x + 3z(1 - x^2)$ and $L_y = \partial_y + 2zy$. Consequently the algebra of all such amplitudes over the actual boundary field is the polynomial algebra in M_0, M_1 , both functionally and at each nonzero algebraic parameter.

This is proved in Theorems 12.2 and 13.2. The proof does not use a formal completeness theorem for boundary periods. Paper III enters only in the final formal-period interpretation, which is explicitly local at boundary denominators known not to vanish.

There are three different levels of assertion. The polynomial boundary system is general. Injectivity of the actual comparison requires the explicit hypotheses of Theorem 8.2. Those hypotheses are verified for the fixed elliptic chain above, for arbitrary polynomial amplitudes. Multiple chains and multiple primitive blocks require additional comparisons; the abstract local theorem Theorem 14.2 does not itself supply their geometric hypotheses.

For the retained geometric results, the phase is real on the real chain. The phase-density, face-lattice, rooted-orbit, and Bonnet arguments are independent of the numerical specialization proof. The latter uses the actual boundary system and the two-dimensional affine orbit, not an unspecified completed current.

2 Polynomial representatives and literal boundary forcing

Put $k = \overline{\mathbb{Q}}$ and $F_0 = k(z)$. Let D be a finite algebraic-linear combination of pushforwards of $[0, 1]^2$ by polynomial maps with real algebraic coefficients. Let $q \in k[x, y]$. For a polynomial two-form ω , write

$$I_D(\omega) = \int_D e^{zq} \omega.$$

When coefficients of ω lie in F_0 , interpret this as a meromorphic function of z by linearity. For each oriented face parametrization $\phi_e : [a_e, b_e] \rightarrow \mathbb{A}^2$, put $p_e = q \circ \phi_e$. The actual boundary field $K_{\partial,0}$ is the differential field over F_0 generated by all functions

$$\int_{a_e}^{b_e} h(t) e^{z p_e(t)} dt, \quad h \in k[t],$$

and the exponential functions at their algebraic endpoints. Only the fixed faces are used; this does not mean the field of all compact line periods.

Let d differentiate only in x, y , and set

$$d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x,y]/F_0}^2 / d_z \Omega_{F_0[x,y]/F_0}^1.$$

Theorem 2.1 (Polynomial-representative boundary system). *The space H_q^2 is finite dimensional. There are forms $\omega_i = r_i(x, y) dx \wedge dy$, with $r_i \in k[x, y]$, whose classes form an F_0 -basis, a matrix $A \in M_n(F_0)$, and polynomial-in- x, y one-forms $\beta_i \in \Omega_{F_0[x,y]/F_0}^1$ such that*

$$q\omega_i = \sum_j A_{ij}(z)\omega_j + d_z\beta_i. \quad (2.1)$$

The actual integral column $U_i = I_D(\omega_i)$ satisfies

$$U' = AU + b, \quad b_i = \int_{\partial D} e^{zq} \beta_i \in K_{\partial,0}. \quad (2.2)$$

Every polynomial two-form ω has an identity

$$\omega = \sum_i a_i(z)\omega_i + d_z\beta, \quad (2.3)$$

where $a_i \in F_0$ and β is polynomial in x, y . Consequently

$$I_D(\omega) = \sum_i a_i(z)U_i + \int_{\partial D} e^{zq} \beta. \quad (2.4)$$

For any finite set of amplitudes, these functions belong to a finite closed rational first-order system whose coordinates are actual E -functions. No spatial denominator is introduced. Parameter denominators may occur.

Proof. Algebraic direct image of the rank-one connection $d + d(zq)$ on $\mathbb{A}^2 \times \mathbb{A}_z^1$ is holonomic. The generic top relative de Rham cohomology is therefore finite dimensional over $k(z)$. Equivalently, it is the generic localization of the Fourier transform of the polynomial Gauss–Manin system. The polynomial complex and Fourier convention are spelled out in [12, Proposition 9.1 and Remarks 9.4], with Fourier parameter $\tau = -z$. This finiteness statement does not identify the rank with a Milnor number without an additional hypothesis.

The monomial two-forms with coefficients in k span the quotient over F_0 , so a finite subset is a basis. The connection on the quotient is

$$\nabla_z[\omega] = [\partial_z\omega + q\omega],$$

because

$$[\partial_z + q, d + z dq \wedge] = dq \wedge - dq \wedge = 0.$$

Expressing $q\omega_i$ in the basis proves (2.1); expressing an arbitrary class proves (2.3).

For a one-form polynomial in the spatial variables, there is the literal identity

$$e^{zq}d_z\beta = d_{x,y}(e^{zq}\beta).$$

Stokes on the original parametrized compact chain proves (2.2) and (2.4). Each pullback of β to a face is a finite F_0 -linear combination of polynomial one-forms in its parameter. Thus every forcing term is an actual face integral with coefficients in F_0 .

To see finiteness on the boundary, if $\deg p = d \geq 2$, division by $\partial_t + zp'(t)$ reduces every polynomial amplitude to degree at most $d - 2$. The leading degree of $(\partial_t + zp')h$, for $h \neq 0$, is $\deg h + d - 1$, with nonzero leading coefficient in F_0 . The remainder is therefore unique. Integration of the exact part gives only endpoint exponentials. Multiplication by p and the same division give a rational first-order system for the reduced moments and the endpoint exponentials. For $d = 1$ the twisted operator is surjective on polynomials; for $d = 0$ ordinary polynomial antiderivatives suffice. In both cases only endpoint terms remain. For finitely many faces, this supplies a finite boundary vector; its fraction field is $K_{\partial,0}$.

The integral coordinates are entire and have algebraic Taylor coefficients. After pullback to the square they have the form $\int_{[0,1]^2} P(u,v) e^{zQ(u,v)} du dv$. For the first m Taylor coefficients, one denominator of the form

$$d^{m+1} \text{lcm}(1, \dots, Cm + C_0)^2$$

suffices, with fixed integers d, C, C_0 ; all conjugates grow exponentially. The same argument in one variable handles the faces. Together with the constructed differential systems, these are the E -function conditions. $\square$

Proposition 2.2 (Actual affine comparison under a change of representatives). *Suppose $\tilde{\omega} = T(z)\omega + d_z\gamma$, with $T \in \text{GL}_n(F_0)$. Put $g = \int_{\partial D} e^{zq}\gamma$. Then the actual columns satisfy*

$$\tilde{U} = TU + g,$$

and their systems are related by

$$\tilde{A} = (T' + TA)T^{-1}, \quad \tilde{b} = Tb + g' - \tilde{A}g.$$

In particular, this is a comparison of the actual affine solution objects, not a statement only about their homogeneous composition factors.

Proof. The first equality is Stokes, and differentiating it proves the other two. If two primitives give the same form identity, their difference is d_z -closed and its integral over ∂D is zero by the same Stokes identity. Thus primitive choices do not introduce an arbitrary constant. $\square$

After adjoining complex constants, write $F = \mathbb{C}(z)$, $K_{\partial} = \mathbb{C}K_{\partial,0}$, and work in a field $\mathcal{M}$ of meromorphic germs containing the actual functions. Integration modulo the actual boundary field gives a canonical differential map

$$K_{\partial} \otimes_{F_0} H_q^2 \longrightarrow \mathcal{M}/K_{\partial}, \quad [\omega] \longmapsto I_D(\omega) \bmod K_{\partial}. \quad (2.5)$$

Here $\mathcal{M}/K_{\partial}$ is used as a differential vector space, not as a quotient field or ring. This map need not be injective for arbitrary chains. The next criterion proves injectivity rather than building it into a completion definition.

3 Polynomial flags and phase densities

3.1 Polynomial flag chains

Definition 3.1 (Polynomial two-chain). A polynomial two-chain over $k_{\mathbb{R}}$ is a finite k -linear combination of chains

$$D = \Phi_*[0, 1]^2, \quad \Phi \in k_{\mathbb{R}}[u, v]^2.$$

Its boundary is the alternating sum of the four polynomial face chains. A polynomial flag chain is a polynomial two-chain for which, after an affine change of the square coordinates,

$$\Phi(u, v) = (X(u), A(u) + vB(u))$$

with $X, A, B \in k_{\mathbb{R}}[u]$.

The chain language is preferable to set-theoretic domains: it records orientation, multiplicity, and algebraic change of variables without requiring injectivity of Φ .

For a polynomial two-form Ω , define

$$F_{D,q,\Omega}(z) = \int_D e^{zq} \Omega.$$

Pulling back to the square writes this as

$$F_{D,q,\Omega}(z) = \int_0^1 \int_0^1 P(u, v) e^{zQ(u,v)} du dv, \quad P, Q \in k[u, v]. \quad (3.1)$$

3.2 Gelfand–Leray density

Let $q \in k_{\mathbb{R}}[x, y]$ be nonconstant and let D be a real polynomial flag chain. Put $T = q(|D|) \subset \mathbb{R}$. Remove the critical values of q , the critical values of the restrictions of q to the faces, and the phase values of the face vertices. The complement is a finite union of open intervals e .

For $t \in e$, let

$$C_t = q^{-1}(t), \quad B_t = C_t \cap |\partial D|, \quad \gamma_t = D \cap C_t.$$

The last object is an oriented relative one-cycle in (C_t, B_t) , with the multiplicities inherited from D .

Let Ω be a polynomial two-form. On the smooth locus of q , choose a one-form α satisfying

$$dq \wedge \alpha = \Omega. \quad (3.2)$$

The restriction class of α to C_t is independent of the choice. Define

$$\rho_e(t) = \int_{\gamma_t} \alpha_t. \quad (3.3)$$

Lemma 3.2 (Phase-density formula). *After discarding chain components on which the pullback of Ω is zero, there is an L^1 density ρ on the compact set $T = q(|D|)$ such that*

$$F_{D,q,\Omega}(z) = \int_T e^{zt} \rho(t) dt. \quad (3.4)$$

On every regular interval e , the restriction ρ_e is real analytic and is a branch of a period of the algebraic Gauss–Manin system. In particular, it admits multivalued holomorphic continuation on the regular phase line. No claim that ρ_e is an algebraic function is made.

Proof. On a compact subinterval of e , the map q is a submersion on the sliced chain. The Gelfand–Leray or coarea formula gives

$$\int_{D \cap q^{-1}(e)} e^{zq} \Omega = \int_e e^{zt} \left(\int_{\gamma_t} \alpha_t \right) dt.$$

The inner integral is a period of the relative algebraic de Rham class $[\alpha_t]$ and therefore is real analytic on e and has the usual multivalued Gauss–Manin continuation. The coarea decomposition of the finite measure obtained by integrating the polynomial density on the compact chain shows that the resulting density is integrable across the finitely many critical and boundary values. Summing the regular intervals proves (3.4). A full-dimensional component on which q is constant cannot occur for nonconstant polynomial q ; a lower-dimensional component pulls every two-form back to zero. $\square$

3.3 Laplace uniqueness

Lemma 3.3 (Compact-support Laplace uniqueness). *Let ρ be a compactly supported integrable function on $\mathbb{R}$. If*

$$\int_{\mathbb{R}} e^{zt} \rho(t) dt = 0 \quad (z \in \mathbb{C}),$$

then $\rho = 0$ almost everywhere. If ρ is analytic on each member of a finite interval decomposition, it vanishes identically on every open interval.

Proof. Restriction to $z = is$, $s \in \mathbb{R}$, is the Fourier transform of the compactly supported L^1 function ρ . Fourier uniqueness gives almost-everywhere vanishing, and analyticity gives intervalwise vanishing. $\square$

For multiphase families it is useful to retain the Cauchy-transform form.

Lemma 3.4 (Cauchy jump separation). *Let $T \subset \mathbb{R}$ be a finite union of compact intervals and let ρ_j be piecewise analytic integrable densities on T . If*

$$\sum_j c_j \int_T e^{zt} \rho_j(t) dt = 0 \quad (z \in \mathbb{C}),$$

then

$$\sum_j c_j \rho_j(t) = 0$$

on every open regular interval.

Proof. All moments of the combined density vanish. Its Cauchy transform

$$\mathcal{C}(s) = \int_T \frac{\sum_j c_j \rho_j(t)}{s - t} dt$$

therefore vanishes for large s and hence on $\mathbb{C} \setminus T$. At an interior point of a regular interval, the two lateral values differ by $2\pi i$ times the analytic density. The jump is zero, so the density vanishes. $\square$

Remark 3.5 (Why real phase support is imposed). For a general complex-valued polynomial phase on a real two-chain, the pushforward is a two-dimensional measure in the complex phase plane. Its holomorphic moments can all vanish without the measure vanishing. Thus the one-variable Cauchy jump argument is unavailable. The phase-fibered hypothesis is mathematical content, not a notational convenience.

4 The face lattice and the generalized Jacobian

Let $\bar{f} : \bar{\mathcal{C}} \rightarrow S$ be a smooth proper family of connected curves and let $\pi : B \rightarrow S$ be a nonempty reduced divisor finite étale over S . Define

$$\mathcal{F}_B = \ker \left(\pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \right). \quad (4.1)$$

Proposition 4.1 (Relative homology sequence). *There is a natural exact sequence of integral local systems*

$$0 \longrightarrow R_1 \bar{f}_* \mathbb{Z} \longrightarrow R_1 \bar{f}_* (\bar{\mathcal{C}}, B; \mathbb{Z}) \xrightarrow{\partial} \mathcal{F}_B \longrightarrow 0. \quad (4.2)$$

Proof. Apply the homology long exact sequence of the pair $(\bar{\mathcal{C}}_s, B_s)$. Connectedness gives $H_0(\bar{\mathcal{C}}_s, \mathbb{Z}) = \mathbb{Z}$, and the map $H_0(B_s, \mathbb{Z}) \rightarrow H_0(\bar{\mathcal{C}}_s, \mathbb{Z})$ is the degree map. The construction is locally constant in s and therefore yields (4.2). $\square$

Proposition 4.2 (Generalized-Jacobian torus). *Let $J = \text{Pic}^0(\bar{\mathcal{C}}/S)$ and $J_B = \text{Pic}^0(\bar{\mathcal{C}}/S, B)$. There is an exact sequence*

$$1 \longrightarrow T_B \longrightarrow J_B \longrightarrow J \longrightarrow 0, \quad T_B = \text{Res}_{B/S} \mathbb{G}_m / \mathbb{G}_m, \quad (4.3)$$

and

$$X^*(T_B) \simeq \mathcal{F}_B. \quad (4.4)$$

Proof. For reduced modulus, the generalized Jacobian is the extension of the ordinary Jacobian by the quotient of the Weil restriction of $\mathbb{G}_m$ by the diagonal $\mathbb{G}_m$; see [13, Chapter V]. Characters are contravariant. The character sequence associated to

$$1 \rightarrow \mathbb{G}_m \rightarrow \text{Res}_{B/S} \mathbb{G}_m \rightarrow T_B \rightarrow 1$$

is

$$0 \rightarrow X^*(T_B) \rightarrow \pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \rightarrow 0,$$

which is (4.4). $\square$

Remark 4.3 (Variance and weights). The relative-homology quotient $\mathcal{F}_B$ is a lattice of boundary classes. The same lattice is the character group of the torus T_B ; the torus itself is dual. In mixed-Hodge terms, relative homology and punctured-curve cohomology carry opposite Tate twists. The slogan “the face module is the toric part” is correct only after this variance is specified.

For a polynomial flag slice, B_t is the finite set of intersections with the oriented faces and

$$\partial \gamma_t \in \mathcal{F}_{B,t}.$$

The equation $\deg \partial \gamma_t = 0$ is simultaneously the balanced face relation and the boundary-of-boundary identity.

5 Rooted Picard–Lefschetz observability

Let H be a finite-dimensional vector space and suppose the monodromy group G is generated by relative Picard–Lefschetz transformations

$$(T_i - I)\xi = \varepsilon_i \langle \tilde{\nu}_i, \xi \rangle \nu_i, \quad \varepsilon_i \in \{\pm 1\}. \quad (5.1)$$

Here ν_i and $\tilde{\nu}_i$ are vanishing and dual vanishing cycles. We use the standard relative Picard–Lefschetz formula; see, for example, [7, Chapters 2–3]. Let γ be a relative slice class.

Definition 5.1 (Rooted variation graph). Put

$$I_\gamma = \{i : \langle \tilde{\nu}_i, \gamma \rangle \neq 0\}.$$

Put a directed edge $i \rightarrow j$ when

$$\langle \tilde{\nu}_j, \nu_i \rangle \neq 0.$$

Let R_γ be the vertices reachable from I_γ , and set

$$\mathcal{V}_\gamma = \text{span}\{\nu_i : i \in R_\gamma\}, \quad \mathcal{O}_\gamma = \text{span}\{g\gamma - \gamma : g \in G\}.$$

Theorem 5.2 (Exact rooted orbit theorem). *One has*

$$\mathcal{O}_\gamma = \mathcal{V}_\gamma.$$

The space is invariant under G .

Proof. The space $\mathcal{O}_\gamma$ is invariant because

$$h(g\gamma - \gamma) = (hg\gamma - \gamma) - (h\gamma - \gamma).$$

If $i \in I_\gamma$, (5.1) gives $\nu_i \in \mathcal{O}_\gamma$. If $\nu_i \in \mathcal{O}_\gamma$ and $i \rightarrow j$, then

$$(T_j - I)\nu_i = \varepsilon_j \langle \tilde{\nu}_j, \nu_i \rangle \nu_j,$$

so $\nu_j \in \mathcal{O}_\gamma$. Hence $\mathcal{V}_\gamma \subseteq \mathcal{O}_\gamma$. The same formula shows that every T_j preserves $\mathcal{V}_\gamma$.

Conversely, for every generator,

$$T_j\gamma - \gamma \in \mathcal{V}_\gamma.$$

If $g\gamma - \gamma \in \mathcal{V}_\gamma$, then

$$T_j g\gamma - \gamma = T_j(g\gamma - \gamma) + (T_j\gamma - \gamma) \in \mathcal{V}_\gamma.$$

The same argument applies to T_j^{-1} , since $T_j^{-1} - I$ has the same rank-one image. Induction on word length in the generators and their inverses gives $g\gamma - \gamma \in \mathcal{V}_\gamma$ for every g , proving the reverse inclusion. $\square$

On absolute curve homology the intersection graph is undirected.

Corollary 5.3 (Connected-graph observability). *Suppose the vanishing cycles span V , their intersection graph is connected, and $I_\gamma \neq \emptyset$. Then*

$$\mathcal{O}_\gamma = V.$$

Proof. Every vertex is reachable from the nonempty active set. Apply Theorem 5.2. $\square$

Proposition 5.4 (Absolute irreducibility). *Assume in addition that V carries a nondegenerate alternating pairing and that the T_i are the associated symplectic transvections. Under the hypotheses of Corollary 5.3, the representation on V is absolutely irreducible.*

Proof. Let $K/\mathbb{Q}$ be a characteristic-zero extension and let $0 \neq W \subset V_K$ be invariant. Choose $0 \neq w \in W$. Since the vanishing cycles span and the pairing is nondegenerate, some $\langle w, \nu_i \rangle$ is nonzero. Then $(T_i - I)w$ is a nonzero multiple of ν_i , so $\nu_i \in W$. Connectivity propagates membership to every vanishing cycle. Thus $W = V_K$. $\square$

6 Denominator-free relative exactness

Definition 6.1 (Admissible observable flag datum). An admissible observable flag datum is a triple (D, q, Ω) with D a polynomial flag chain, $q \in k_{\mathbb{R}}[x, y]$, and $\Omega \in \Omega_{k[x, y]/k}^2$, satisfying:

- (A1) q has finite critical locus and every fibre $q^{-1}(t)$ is connected;
- (A2) the generic affine fibre has one place at infinity;
- (A3) after resolving the compactified pencil, every finite singular fibre has exactly one ordinary node and the finite critical values are distinct; over their complement the fibres form a smooth proper curve family, and the face intersections become a finite étale divisor after a finite base change;
- (A4) the resulting Lefschetz vanishing cycles span H_1 of the compactified generic fibre and have connected intersection graph;
- (A5) the sliced chain is nonzero on at least one regular phase interval, and on every interval on which it is nonzero it has nonzero incidence with at least one Lefschetz vanishing cycle.

6.1 Bonnet's theorem

For a polynomial map $q : \mathbb{C}^2 \rightarrow \mathbb{C}$, a polynomial one-form ω is *topologically relatively exact* if its restriction to every generic fibre has zero integral on every loop. It is *algebraically relatively exact* if

$$\omega = dR + a dq, \quad R, a \in \mathbb{C}[x, y].$$

Lemma 6.2 (Bonnet's hypotheses in relative dimension one). *Let $q : \mathbb{C}^2 \rightarrow \mathbb{C}$ be nonconstant. If its critical locus is finite and every fibre is connected, then q is primitive and quasi-fibered in Bonnet's terminology.*

Proof. A finite critical locus has codimension two, so q is nonsingular in codimension one. Since the target has dimension one, Bonnet's requirement that the fibres be 1-generically connected means that every fibre is connected. His 2-generic nonemptiness condition is automatic: for every $c \in \mathbb{C}$, the proper ideal $(q - c) \subset \mathbb{C}[x, y]$ has a zero by the Nullstellensatz. The weaker generic connectedness and nonemptiness conditions defining primitivity follow as well. $\square$

Theorem 6.3 (Bonnet–Gavrilov relative exactness). *If $q : \mathbb{C}^2 \rightarrow \mathbb{C}$ has finite critical locus and every fibre is connected, then every topologically relatively exact polynomial one-form is algebraically relatively exact.*

Proof. Apply Lemma 6.2 and [3, Theorem 1.5]. Bonnet also records the earlier two-variable connected-and-reduced theorem of Gavrilov. $\square$

6.2 From a zero phase density to relative exactness

Let Ω be a polynomial two-form. Choose a polynomial one-form ω with

$$d\omega = \Omega; \quad (6.1)$$

this is possible because $H_{\text{dR}}^2(\mathbb{A}^2) = 0$. On a regular fibre choose α as in (3.2). The Gelfand–Leray formula is

$$\nabla[\omega_t] = [\alpha_t]. \quad (6.2)$$

Lemma 6.4 (Observable density kills the Gelfand–Leray class). *Let (D, q, Ω) satisfy (A2)–(A5) of Definition 6.1. If the phase density vanishes on a regular interval on which the sliced chain is nonzero and has nonzero Lefschetz incidence, then*

$$[\alpha_t] = 0 \in H_{\text{dR}}^1(C_t)$$

for the generic affine fibre.

Proof. On the active interval in the hypothesis, the period $\int_{\gamma_t} \alpha_t$ vanishes and hence, by analytic continuation, on every continuation of that branch. Therefore $[\alpha_t]$ annihilates

$$\text{span}\{g\gamma_t - \gamma_t : g \in G\}.$$

By Corollary 5.3, this is the complete projective weight-one homology. The one-place-at-infinity hypothesis identifies this with $H_1(C_t)$ of the affine fibre. The de Rham–Betti pairing is perfect, so $[\alpha_t] = 0$. $\square$

Lemma 6.5 (The primitive one-form is fibrewise exact). *Under the hypotheses of Lemma 6.4, the restriction of ω to the generic fibre is exact.*

Proof. Equation (6.2) and Lemma 6.4 show that $[\omega_t]$ is horizontal. It is therefore monodromy invariant. For a Picard–Lefschetz transvection, invariance of the cohomology class forces its period on the corresponding vanishing cycle to be zero. Since the vanishing cycles span $H_1(C_t)$, all periods of ω_t vanish. Hence the restriction is exact. $\square$

Theorem 6.6 (Polynomial relative Stokes). *Let (D, q, Ω) be admissible observable. If the phase density vanishes on every regular interval, there is a $a \in k[x, y]$ such that*

$$\Omega = da \wedge dq. \quad (6.3)$$

Consequently

$$e^{zq}\Omega = d(e^{zq}a \, dq). \quad (6.4)$$

Proof. By (A5) there is an active regular interval with nonzero Lefschetz incidence. The density vanishes there by hypothesis, so Lemmas 6.4 and 6.5 show that ω is topologically relatively exact. Apply Theorem 6.3:

$$\omega = dR + a \, dq$$

with $R, a \in \mathbb{C}[x, y]$. Taking exterior derivatives and using (6.1) gives (6.3). The coefficients descend to k : after fixing the finitely many monomials occurring in one solution, coefficient comparison is a finite linear system over k . Finally, $dq \wedge dq = 0$ gives (6.4). $\square$

Theorem 6.7 (Single-phase denominator-free functional comparison). *For an admissible observable datum, $I_D(\Omega) \equiv 0$ implies $\Omega = da \wedge dq$ for some $a \in k[x, y]$. Thus the zero relation is reduced by polynomial Stokes to the compact polynomial-line faces.*

Proof. Functional vanishing and Lemma 3.3 make every regular phase density zero. Apply Theorem 6.6 and ordinary Stokes:

$$F(z) = \int_{\partial D} e^{zq} a \, dq.$$

Every face is polynomially parametrized, so the right side is a finite combination of compact polynomial-line functions. It vanishes identically because $F \equiv 0$. The functional formal theorem of [10] makes this boundary relation formal. $\square$

Remark 6.8 (Why one place at infinity appears). If the generic affine fibre has $r > 1$ points at infinity, then $H_1(C_t)$ contains an additional rank- $r-1$ puncture lattice. Projective Picard–Lefschetz observability does not detect it. One may replace (A2) by an explicit infinity-lattice observability hypothesis, but it cannot simply be omitted.

7 Hodge rigidity and Jacobian correspondences

Theorem 7.1 (Deligne uniqueness for irreducible factors). *An irreducible complex local system underlying a polarizable complex variation of Hodge structure carries such a variation uniquely up to tensoring by a rank-one complex Hodge structure $\mathbb{C}(r, s)$, equivalently up to a shift of the Hodge bigrading.*

This is [5, Proposition 1.13]; see also the convenient formulation in [1, Theorem 2].

Theorem 7.2 (Weight-one Hodge rigidity). *Let $\mathbb{V}$ and $\mathbb{W}$ be nonzero polarizable rational variations of Hodge structure of effective weight one on a connected smooth algebraic base. Assume their complex local systems are irreducible. Every nonzero flat map*

$$\phi : \mathbb{V}_{\mathbb{C}} \longrightarrow \mathbb{W}_{\mathbb{C}}$$

is a morphism of complex variations of Hodge structure.

Proof. Irreducibility makes ϕ an isomorphism of complex local systems. Transport the Hodge bigrading of $\mathbb{W}$ to $\mathbb{V}$. By Theorem 7.1, the transported grading differs from the original one by a shift $\mathbb{C}(r, s)$. Because both variations are effective of weight one and arise from rational Hodge structures, both types $(1, 0)$ and $(0, 1)$ occur. The shifted set of types can again be exactly $\{(1, 0), (0, 1)\}$ only when $r = s = 0$. Thus ϕ preserves the Hodge bigrading and hence the Hodge filtration. $\square$

Remark 7.3 (The reducible warning). The theorem fails for repeated reducible local systems. A constant elliptic curve gives a two-dimensional trivial local system; most constant matrices are flat but do not preserve $H^{1,0}$. The irreducibility hypothesis is therefore load-bearing.

Proposition 7.4 (Algebraic realization). *Let $f_i : \mathcal{C}_i \rightarrow S$ and $f_j : \mathcal{C}_j \rightarrow S$ be smooth proper curve families over a smooth connected complex algebraic base, with relative Jacobians J_i, J_j . A rational morphism of weight-one variations*

$$R^1 f_{i*} \mathbb{Q} \longrightarrow R^1 f_{j*} \mathbb{Q}$$

is induced, after multiplication by a positive integer and with the usual contravariance convention, by a homomorphism of abelian schemes between J_i and J_j . Its action on R^1 is represented by an algebraic divisor correspondence on $\mathcal{C}_i \times_S \mathcal{C}_j$.

Proof. The Hodge realization functor from abelian schemes over S to torsion-free polarizable integral variations of types $(-1, 0)$ and $(0, -1)$ is fully faithful; see [6, Rappel (4.4.3)]. After multiplying the rational morphism by an integer, full faithfulness therefore supplies the corresponding homomorphism of relative Jacobians.

The homomorphism and the Poincaré bundle determine a relative line bundle on $J_i \times_S J_j$. Pulling it back along the Abel–Jacobi maps (after an inessential finite base change if relative degree-one divisor classes are needed) gives a line bundle on $\mathcal{C}_i \times_S \mathcal{C}_j$. Its first Chern class, represented by a relative divisor with rational coefficients after rescaling, acts on R^1 as the original morphism. $\square$

Corollary 7.5 (Complex intertwiners are generated by correspondences). *Under the hypotheses of Theorem 7.2, every complex flat intertwiner is a complex linear combination of rational VHS morphisms and therefore, by Proposition 7.4, of algebraic Jacobian correspondences.*

Proof. For rational local systems, the intertwiner space is the kernel of a finite system of rational linear equations. Hence scalar extension gives

$$\mathrm{Hom}_{\pi_1}(\mathbb{V}, \mathbb{W}) \otimes_{\mathbb{Q}} \mathbb{C} \simeq \mathrm{Hom}_{\pi_1}(\mathbb{V}_{\mathbb{C}}, \mathbb{W}_{\mathbb{C}}).$$

Apply Theorem 7.2 and Proposition 7.4 to the rational summands. $\square$

7.1 Disjoint blocks and the remaining correspondence obligation

Proposition 7.6 (Disjoint-block comparison on a fixed common base). *Let a finite family of admissible data be pulled to a fixed common phase base on which the face points are sections. Suppose the absolute complex local systems on this base are irreducible and pairwise nonisomorphic, and the actual slice-variation space projects onto each active factor. A constant-coefficient functional relation then reduces termwise by polynomial relative Stokes to a compact polynomial-line boundary relation.*

Proof. Laplace uniqueness gives a zero combined density on each active interval. Its monodromy differences annihilate the actual variation submodule of the direct sum of absolute systems. A submodule of a sum of pairwise nonisomorphic simple modules is the sum of some factors; surjectivity to every active factor makes it the whole sum. Each scaled Gelfand–Leray class is zero. The proof of Theorem 6.6 applies to each scaled form, and the residual relation is on the faces. $\square$

The hypotheses refer to the chosen common base, not to a representation before an unspecified finite cover. For repeated isomorphic blocks, absolute Hodge intertwiners do not determine the relative slice extension. Moreover a generic Jacobian correspondence need not preserve polynomial representatives without phase denominators. A general isotypic formal comparison requires an actual extension comparison and a compatible polynomial or local lattice realization. We do not infer either from Hodge rigidity alone. For the elliptic theorem below, these issues are replaced by a direct actual-column proof.

8 An actual-solution comparison criterion

Let $\mathcal{B}$ be a finite rational boundary differential system, let L_∂/F be a Picard–Vessiot field for that system containing its actual solution coordinates, and let $\mathcal{N} \subset L_\partial$ be their finite-dimensional F -linear differential span. Include 1 when needed and put $K_\partial = F(\mathcal{N}) \subset L_\partial$. All fields are embedded in a common no-new-constants differential overfield. Let $\mathcal{T}_\partial$ be the tensor category generated by $\mathcal{B}$, allowing subquotients and duals. Its objects are precisely the differential modules trivialized by L_∂ [14, Theorem 1.5.2].

Lemma 8.1 (Differentially finite elements of a Picard–Vessiot field). *Every finite-dimensional F -differential subspace of L_∂ belongs to $\mathcal{T}_\partial$.*

Proof. Let R be the Picard–Vessiot ring. If $f \in L_\partial$ satisfies a scalar linear equation over F , all its Galois translates satisfy that equation. Their constant span is finite dimensional. After trivializing the torsor, a rational function with finite-dimensional translation orbit has no pole: the union of the polar loci of a finite basis is a proper closed subset invariant under all translations, whereas the group acts transitively on itself. This remains valid component by component if the group is disconnected. Thus $f \in R$; see the locally finite argument in [14, proof of Proposition 1.3.33].

Write $R = F[Y_{ij}, (\det Y)^{-1}]$ for a boundary fundamental matrix. A bounded span of monomials in Y_{ij} and $(\det Y)^{-1}$ is stable under differentiation, since $Y' = A_\partial Y$ and $((\det Y)^{-1})' = -\operatorname{tr}(A_\partial)(\det Y)^{-1}$. Such a span is a quotient of finite sums of tensor constructions in the boundary module and its determinant dual. Every finite-dimensional F -differential subspace of R lies in one of these spans. Closure under subquotients and the Picard–Vessiot tensor equivalence [14, Theorem 1.5.2] prove the claim. $\square$

Theorem 8.2 (Actual one-column comparison). *Suppose an actual column U satisfies $U' = AU + b$, with $b_i \in \mathcal{N}$, and $n \geq 2$. Assume:*

1. *the homogeneous Galois group over F contains SL_n ;*
2. *its adjoint module $\operatorname{End}^0(A)$ does not belong to $\mathcal{T}_\partial$;*
3. *at least one coordinate of U is not in $\mathcal{N}$.*

Then $U_1, \dots, U_n$ are algebraically independent over $K_\partial \subset L_\partial$. In the setting of Theorem 2.1, (2.5) is injective, and its image is the span of the actual polynomial-amplitude periods modulo K_∂ .

Proof. Let $E = \mathcal{N} + \sum_i F U_i$. The differential module $E/\mathcal{N}$ is a quotient of the coefficient module associated with A . Use the coefficient module with basis e_i and $\partial e_i = \sum_j A_{ij} e_j$; the map $e_i \mapsto U_i \bmod \mathcal{N}$ is a differential map. Its horizontal representation is dual to the solution representation of $Y' = AY$, so it is irreducible and has the same adjoint tensor module. By the third assumption, the quotient is nonzero and hence is that entire irreducible module.

If all U_i belonged to L_∂ , Lemma 8.1 would put E , then $E/\mathcal{N}$, and then its adjoint module in $\mathcal{T}_\partial$, contradicting the second assumption. Thus $U \notin L_\partial^n$.

In a joint Picard–Vessiot extension for A and the boundary system, the kernel of restriction to L_∂ maps to a normal subgroup H of the homogeneous group $G \subset \operatorname{GL}_n$. A normal subgroup of a group containing SL_n either contains SL_n or has trivial projective image. Indeed, if $H \cap \operatorname{SL}_n$ is not all of SL_n , it is finite central. For each $h \in H$, the commutator map from the connected group SL_n to this finite intersection is constant and equals the identity at 1. Thus h centralizes SL_n and is scalar. This also excludes a hidden noncentral finite projective image. In the latter, projectively

trivial case, H acts trivially on $\text{End}^0(A)$, so the adjoint module is trivialized by L_∂ , a contradiction. Therefore the homogeneous group after base change to L_∂ , and hence to K_∂ , contains SL_n .

Let Y be a fundamental matrix over a joint field for the inhomogeneous system over K_∂ . Its affine Galois group is a subgroup of $\mathbb{C}^n \rtimes G_{K_\partial}$, acting by $\sigma(U) = U + Yc_\sigma$. Its translation kernel is an invariant vector subspace, hence either 0 or $\mathbb{C}^n$. If it is $\mathbb{C}^n$, translations make the orbit of U Zariski dense in $\mathbb{A}^n$. If it is zero, the remaining regular algebraic cocycle of the reductive homogeneous group is a coboundary. Here reductivity follows also directly: a unipotent radical would lie in SL_n and be normal under SL_n , which has no nontrivial connected normal unipotent subgroup. Thus

$$U = a + Yc, \quad a \in K_\partial^n, \quad c \in \mathbb{C}^n.$$

The vector c is nonzero, since otherwise $U \in K_\partial^n$. The SL_n -orbit of a nonzero vector is $\mathbb{A}^n \setminus \{0\}$, again dense. In either case a polynomial over K_∂ vanishing at U vanishes on a dense affine orbit and is zero. Linear independence modulo the boundary field, and then the last assertion by normal-form reduction, follow immediately. $\square$

Remark 8.3. The third condition concerns only rational *linear* combinations of a finite actual boundary vector. It is strictly weaker than the desired nonmembership in the boundary fraction field. The proof is what upgrades this linear test to the fraction-field statement. A nonzero abstract cohomology class alone would not suffice.

9 The elliptic polynomial normal form

Fix the counterclockwise triangle

$$D = \text{conv}\{(-1/2, -1), (1/2, 0), (-1/2, 1)\} = \{-1/2 \leq x \leq 1/2, |y| \leq 1/2 - x\},$$

and

$$q = y^2 - x^3 + 3x, \quad M_j = \int_D x^j e^{zq} dx dy \quad (j = 0, 1).$$

A polynomial parametrization is $\Phi(u, v) = (u - 1/2, (1 - u)(2v - 1))$, with Jacobian $2(1 - u)$.

For $\beta = P dy - Q dx$, one has

$$dz\beta = (L_x P + L_y Q) dx \wedge dy, \quad L_x = \partial_x + 3z(1 - x^2), \quad L_y = \partial_y + 2zy.$$

The operators act in separate variables. Their one-variable quotients have bases $1, x$ and 1 , respectively; both operators are injective on polynomials. Tensoring the two one-variable complexes proves

$$H_q^2 = F_0[dx \wedge dy] \oplus F_0[x dx \wedge dy]. \quad (9.1)$$

In particular the rank is exactly two, not merely at most two.

The explicit reduction recurrences are

$$[x^i y^j] = -\frac{j-1}{2z} [x^i y^{j-2}] \quad (j \geq 1), \quad (9.2)$$

$$[x^{m+2}] = [x^m] + \frac{m}{3z} [x^{m-1}] \quad (m \geq 0), \quad (9.3)$$

with the zero-coefficient terms omitted. Each use records a polynomial primitive. Hence every $r \in k[x, y]$ has a unique remainder $a_r(z) + b_r(z)x$, with $a_r, b_r \in k[z, z^{-1}]$, and an identity

$$r = a_r + b_r x + L_x P_r + L_y Q_r, \quad P_r, Q_r \in k[z, z^{-1}, x, y]. \quad (9.4)$$

Let

$$C = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -5/6 & 0 \\ 0 & -7/6 \end{pmatrix}, \quad A = C + D_0/z.$$

The following are identities of polynomial forms, with Laurent coefficients in z :

$$q \, dx \wedge dy = -\frac{5}{6z} \, dx \wedge dy + 2x \, dx \wedge dy + d_z \beta_0, \quad (9.5)$$

$$q x \, dx \wedge dy = 2 \, dx \wedge dy - \frac{7}{6z} x \, dx \wedge dy + d_z \beta_1, \quad (9.6)$$

where

$$\beta_0 = \frac{x}{3z} \, dy - \frac{y}{2z} \, dx, \quad \beta_1 = \frac{x^2 - 2}{3z} \, dy - \frac{xy}{2z} \, dx.$$

Define actual face integrals

$$B_{x,j} = \int_{\partial D} x^j e^{zq} \, dy, \quad B_{y,j} = - \int_{\partial D} x^j y e^{zq} \, dx.$$

Then

$$\begin{pmatrix} M_0 \\ M_1 \end{pmatrix}' = \begin{pmatrix} -5/(6z) & 2 \\ 2 & -7/(6z) \end{pmatrix} \begin{pmatrix} M_0 \\ M_1 \end{pmatrix} + \frac{1}{z} \begin{pmatrix} B_{y,0}/2 + B_{x,1}/3 \\ B_{y,1}/2 + B_{x,2}/3 - 2B_{x,0}/3 \end{pmatrix}. \quad (9.7)$$

This is obtained by integrating (9.6), not by choosing an isomorphic abstract connection and then normalizing a solution.

9.1 A finite actual boundary field

Put

$$p(x) = -x^3 + x^2 + 2x + \frac{1}{4}, \quad P_j(z) = \int_{-1/2}^{1/2} x^j e^{zp(x)} \, dx, \quad Q_0(z) = \int_{-1}^1 e^{z(y^2 - 11/8)} \, dy,$$

and $E_a = e^{-3z/8}$, $E_b = e^{11z/8}$. Parametrizing the three oriented edges gives

$$B_{x,j} = 2P_j - (-1/2)^j Q_0, \quad B_{y,j} = P_j - 2P_{j+1}. \quad (9.8)$$

The one-variable reductions imply

$$K_{\partial,0} = k(z)(P_0, P_1, Q_0, E_a, E_b). \quad (9.9)$$

Indeed the two slanted faces have the same phase p ; their arbitrary polynomial amplitudes reduce to P_0, P_1 and endpoints. The vertical quadratic amplitudes reduce to Q_0 and endpoints.

For completeness, a closed boundary system is

$$P'_0 = \left(\frac{17}{36} - \frac{1}{3z} \right) P_0 + \frac{14}{9} P_1 + \frac{E_b + 5E_a}{18z}, \quad (9.10)$$

$$P'_1 = \left(\frac{28}{27} + \frac{1}{9z} \right) P_0 + \left(\frac{163}{108} - \frac{2}{3z} \right) P_1 + \frac{-53E_b + 41E_a}{108z}, \quad (9.11)$$

$$Q'_0 = \left(-\frac{11}{8} - \frac{1}{2z} \right) Q_0 + \frac{E_a}{z}, \quad E'_a = -\frac{3}{8} E_a, \quad E'_b = \frac{11}{8} E_b. \quad (9.12)$$

For example, $P_2 = (2P_1 + 2P_0)/3 - (E_b - E_a)/(3z)$, so (9.7) closes with this boundary vector. The only finite pole of the combined rational system is zero.

9.2 The actual normalization at zero

The face forcing in (9.7) is $b(z)/z$ with b entire. Writing $M = \sum v_m z^m$, $b = \sum b_m z^m$, its holomorphic solution obeys

$$(mI - D_0)v_m = Cv_{m-1} + b_m, \quad v_{-1} = 0.$$

Every $mI - D_0$, $m \geq 0$, is invertible. Thus there is at most one holomorphic solution at zero with these actual forcing functions. The integral supplies it, with

$$M_0(0) = 1, \quad M_1(0) = -\frac{1}{6}.$$

This supplies a second, independent check that no unspecified homogeneous solution has been added to the actual column.

10 Explicit comparison with elliptic Gauss–Manin periods

The regular fibers are

$$E_t: y^2 = f_t(x) = x^3 - 3x + t.$$

On a locally constant closed cycle, let $h_j(t) = \int x^j dx / (2y)$ for $j = 0, 1$. Set

$$R_0(x, t) = \frac{4 - tx - 2x^2}{6(t^2 - 4)}, \quad R_1(x, t) = \frac{2t - 2x - tx^2}{6(t^2 - 4)}.$$

Direct differentiation of rational forms gives

$$\partial_t \left(\frac{x^j dx}{2y} \right) - \sum_{k=0}^1 G_{jk}(t) \frac{x^k dx}{2y} = d_x \left(\frac{R_j(x, t)}{y} \right), \quad G(t) = \frac{1}{6(t^2 - 4)} \begin{pmatrix} -t & 2 \\ -2 & t \end{pmatrix}. \quad (10.1)$$

These identities can be checked after multiplication by $4y^3$. Integration on closed cycles proves

$$h' = G(t)h, \quad (tI - C)h' = -(D_0 + I)h. \quad (10.2)$$

For the convention $dq \wedge \alpha = x^j dx \wedge dy$, the Gelfand–Leray form is $-x^j dx / (2y)$. The common minus sign does not change the matrix G or the nonvanishing of its periods. The forms $dx / (2y)$, $x dx / (2y)$ are the usual two-dimensional de Rham basis of the elliptic curve (the latter is of the second kind at infinity). The exact rational-form identities identify the actual Gauss–Manin connection, not only its local exponents.

With Laplace convention $\mathcal{L}(tT) = \partial_z \mathcal{L}(T)$ and $\mathcal{L}(\partial_t T) = -z\mathcal{L}(T)$, applying the Weyl substitution to (10.2) gives

$$-\partial_z(zU) - C(-zU) = -(D_0 + I)U, \quad \text{hence} \quad zU' = zCU + D_0U.$$

This is exactly the homogeneous part of (9.7). The polynomial certificates (9.6) give its actual relative forcing. This comparison concerns the displayed polynomial Weyl presentation and its generic transformed connection. It does not identify arbitrary lattices or every possible middle extension from its generic matrix alone. For the application no such identification is needed: the polynomial certificates already give the actual system, and its Galois group is computed directly below. No arithmetic normalization is inferred from independently split punctual distributions.

11 The actual column is not linearly generated by the faces

In this section all linear coefficients may be complex. Let

$$\mathcal{N} = \text{span}_{\mathbb{C}(z)}\{1, P_0, P_1, Q_0, E_a, E_b\}.$$

It is a differential space by the displayed boundary system.

Lemma 11.1 (Nonzero actual quotient). *One has $M_0 \notin \mathcal{N}$.*

Proof. Suppose otherwise. Clearing rational denominators yields

$$P(z)M_0(z) = \sum_j R_j(z)g_j(z), \quad 0 \neq P \in \mathbb{C}[z], \quad R_j \in \mathbb{C}[z], \quad g_j \in \{1, P_0, P_1, Q_0, E_a, E_b\}.$$

The inverse Laplace distributions of the actual line periods have algebraic densities on every regular real interval: on a monotone branch of a polynomial phase the density is $h(x(t))/p'(x(t))$. The exponentials and 1 give point masses. Derivatives of these distributions still have algebraic germs on regular intervals. Laplace uniqueness for compactly supported distributions therefore implies that the density germ $\rho(t)$ of M_0 , on a regular interval just to the right of $t = 0$, satisfies

$$P(-\partial_t)\rho(t) \quad \text{is algebraic over } \mathbb{C}(t). \quad (11.1)$$

We verify a nonzero elliptic monodromy variation of this particular germ. Put $f(x) = -x^3 + 3x$, let $r_2(s) \in (-1, 1)$ be its middle inverse branch for $-2 < s < 2$, and let $c(t)$ be the inverse branch of p beginning at $c(0) \in (-1/5, 0)$. Set $a(t) = 1/2 - c(t)$. Coarea on the triangle gives, near zero,

$$\rho(t) = \int_{-a(t)}^{a(t)} \frac{dy}{3(1 - r_2(t - y^2)^2)}. \quad (11.2)$$

Here $p'(x) = -3x^2 + 2x + 2$ is positive on $[-1/5, 1]$. Moreover $p(-1/5) < 0 < p(0)$ and $p(4/5) < 2 < p(1)$, so $c(t)$ is a single real-analytic branch for $0 \leq t \leq 2$, with $c(0) \in (-1/5, 0)$ and $c(2) \in (4/5, 1)$. For $|y| \leq |a(t)|$ one has

$$f(c(t)) = t - a(t)^2 \leq t - y^2 \leq t < 2, \quad f(c(t)) > -2.$$

Thus the middle inverse branch is regular on this moving segment for $0 < t < 2$. Writing the signed integral on the fixed interval as

$$a(t) \int_{-1}^1 \frac{ds}{3(1 - r_2(t - a(t)^2 s^2)^2)}$$

proves real analyticity and hence local holomorphic continuation along that interval. At $t = 11/8$ one has $c(t) = 1/2$, so the endpoints meet, but the displayed expression is analytic and zero there. This specifies the continuation through that otherwise excluded boundary value. Near $t = 2$, $a(t) < 0$ and $a(2) \neq 0$, and both endpoints are unramified.

Write $\varepsilon = 2 - t$ and $r_2(t - y^2) = 1 - v$. Since $f(1 - v) = 2 - 3v^2 + v^3$, near the node the integrand is

$$\frac{1}{2\sqrt{3}\sqrt{\varepsilon + y^2}} + O(1).$$

Choose a fixed $b > 0$ smaller than $|a(2)|$. The outer parts $|y| \geq b$ extend holomorphically to $t = 2$; on $|y| \leq b$ the remainder in the displayed expansion is uniformly bounded for $\varepsilon > 0$. Since the integration orientation is reversed near 2, it follows that

$$\rho(t) = \frac{1}{2\sqrt{3}} \log(2-t) + O(1) \quad (t \rightarrow 2^-). \quad (11.3)$$

The local geometry makes the corresponding variation precise. Put $v = 1 - x$ and $u = v\sqrt{3 - v}$. Then $q - 2 = y^2 - u^2$ near the node, so a nearby local fiber is an annulus. The path outside a small nodal neighborhood and its two endpoints return under a sufficiently small loop. Its variation lies in the one-dimensional cycle space of that annulus. The coefficient in (11.3) proves it is nonzero. The local core cycle δ_+ is fixed by this nodal monodromy (equivalently by its Dehn twist). Consequently

$$(T_+ - I)\rho = c_* h_+(t), \quad c_* \neq 0, \quad h_+(t) = \int_{\delta_+} \frac{dx}{2y}, \quad T_+ h_+ = h_+.$$

This is also the local annulus proof of the Picard–Lefschetz formula in this case; the nonzero coefficient has just been checked on the actual path by (11.2).

A power T_+^m fixes the algebraic germ in (11.1). Analytic continuation commutes with the constant-coefficient operator, so $P(-\partial_t)h_+ = 0$. But every solution germ of a nonzero constant-coefficient differential equation extends to an entire function of t . The elliptic period h_+ does not. Indeed, for $-2 < t < 2$ let $r_1 < r_2 < r_3$ be the roots of $x^3 - 3x + t$. Up to a fixed nonzero factor,

$$h_+(t) = \int_{r_2}^{r_3} \frac{dx}{\sqrt{(x-r_1)(x-r_2)(r_3-x)}}.$$

As $t \rightarrow -2^+$, $r_1, r_2 \rightarrow -1$ and $r_3 \rightarrow 2$. Put $\delta = r_2 - r_1$. On $x = r_2 + v$, $\delta \leq v \leq b$ for a fixed small $b > 0$, the absolute value of the integrand is bounded below by c/v with a fixed $c > 0$. The whole real integral has one sign and is therefore unbounded as $\delta \rightarrow 0$. The analogous upper bound gives logarithmic growth. In particular this continuation cannot be entire. This contradiction proves the lemma. $\square$

Remark 11.2. The singularity at $t = 2$ is a singularity of the analytic continuation of a regular phase-density germ. It is not a claim that the original compactly supported density is nonzero outside $q(D) = [-11/8, 11/8]$.

12 Galois disjointness from the actual boundary system

12.1 The rank-two homogeneous group

Eliminating the second coordinate from $Y' = AY$ gives

$$y_0'' + \frac{2}{z} y_0' + \left(\frac{5}{36z^2} - 4 \right) y_0 = 0.$$

With $w = zy_0$, this is

$$w'' = \left(4 - \frac{5}{36z^2} \right) w.$$

Equivalently $y_0 = z^{-1/2}v(2z)$, where v satisfies the modified Bessel equation of order $1/3$. The Bessel differential-Galois classification states that the group is SL_2 for an order not in $1/2 + \mathbb{Z}$

[14, Example 1.3.32]; an algebraic rank-one twist does not change this connected derived group. Moreover $\text{tr } A = -2/z$, so the determinant solution is a constant multiple of z^{-2} . Thus the homogeneous group of our system is exactly SL_2 . Its adjoint formal exponential factors at infinity are

$$1, e^{4z}, e^{-4z}. \quad (12.1)$$

12.2 The boundary has only one nonabelian simple factor

The homogeneous rank-two block of the boundary system is the cubic moment module for p . Write

$$\alpha = \frac{\sqrt{7}}{3}, \quad \beta = \alpha^3 = \frac{7\sqrt{7}}{27}, \quad c_0 = \frac{107}{108}.$$

Then

$$p(\alpha s + 1/3) = c_0 + \beta(-s^3 + 3s). \quad (12.2)$$

The centered cubic homogeneous moments satisfy

$$V' = \left[\begin{pmatrix} 0 & 2\beta \\ 2\beta & 0 \end{pmatrix} + \frac{1}{z} \begin{pmatrix} -1/3 & 0 \\ 0 & -2/3 \end{pmatrix} \right] V,$$

up to the scalar exponential $e^{c_0 z}$ and an algebraic constant basis change. Its first coordinate is again modified Bessel of order $1/3$, now at $2\beta z$. Its derived group is SL_2 , and its adjoint formal exponential factors are

$$1, e^{4\beta z}, e^{-4\beta z}. \quad (12.3)$$

All other diagonal factors in the full boundary system are rank one: the vertical quadratic factor and the endpoint exponentials. The kernel of the Galois action on these diagonal factors is unipotent, because the system is block triangular. The reductive quotient therefore has one simple derived factor SL_2 , detected by the cubic block; its remaining quotient is diagonalizable.

Lemma 12.1 (Projective exclusion). *The adjoint module of A does not belong to the tensor category of the boundary system.*

Proof. Otherwise that boundary group would have the projective Galois group PGL_2 of A as a quotient. The unipotent radical maps trivially to this reductive quotient, so the quotient factors through the reductive boundary group. Its restriction to the unique simple derived SL_2 is the usual projective quotient up to an automorphism of PGL_2 ; such automorphisms over $\mathbb{C}$ are inner. After aligning this restriction with the cubic block, the two projective maps agree on the whole group: their discrepancy centralizes PGL_2 , whose centralizer in itself is trivial. Thus the two adjoint differential modules would be isomorphic over $\mathbb{C}(z)$.

Their formal exponential factors (12.1) and (12.3) disagree, since $\beta \neq \pm 1$. Such factors are preserved by differential-module isomorphism. This is a contradiction. Rank-one endpoint exponentials do not affect an adjoint module, so they cannot remove the obstruction. $\square$

Theorem 12.2 (Complete actual elliptic comparison). *For the fixed triangle and phase above,*

$$M_0, M_1 \text{ are algebraically independent over } K_\partial.$$

In particular, over the arithmetic boundary field,

$$K_{\partial,0}[I_D(r \, dx \wedge dy) : r \in k[x, y]] = K_{\partial,0}[M_0, M_1] \simeq K_{\partial,0}[X_0, X_1]. \quad (12.4)$$

The canonical map (2.5) is an isomorphism onto the two-dimensional space of actual polynomial-amplitude periods modulo the boundary field. For every polynomial amplitude,

$$I_D(r dx \wedge dy) \in K_\partial \iff [r dx \wedge dy] = 0 \text{ in } H_q^2. \quad (12.5)$$

All such boundary reductions are the polynomial identities (9.4) integrated by Stokes.

Proof. The homogeneous group is SL_2 , Lemma 12.1 gives projective disjointness, and Lemma 11.1 gives the required nonzero actual quotient. Apply Theorem 8.2. The equality of algebras and the equivalence follow from the unique normal form and algebraic independence. The statement over k follows from the stronger statement after adjoining complex constants. $\square$

Corollary 12.3 (Full actual translation kernel). *Over the complex actual boundary field, the affine Picard–Vessiot group of the elliptic column and its homogeneous fundamental matrix is $\mathbb{G}_a^2 \rtimes \mathrm{SL}_2$. In particular, the translation kernel is all of $\mathbb{C}^2$.*

Proof. The homogeneous group after base change is SL_2 , and its invariant translation kernel is either zero or the entire standard representation. If it were zero, the splitting argument in Theorem 8.2 would give $M = a + Yc$ with $a \in K_\partial^2$ and $c \neq 0$. Every element of the actual boundary field is meromorphic at zero, being a ratio of entire functions. Hence $Yc = M - a$ would be a nonzero meromorphic homogeneous solution at zero. Its first Laurent term $z^m v$ would satisfy $mv = D_0 v$ with $m \in \mathbb{Z}$, impossible since the eigenvalues of D_0 are $-5/6$ and $-7/6$. Thus the full translation kernel occurs. A subgroup of $\mathbb{G}_a^2 \rtimes \mathrm{SL}_2$ containing all translations and surjecting to SL_2 is the entire semidirect product. $\square$

13 Numerical comparison and the correct fiber algebra

Lemma 13.1 (Scalar descent for functional relations). *For functions with Taylor coefficients in k , extending constants from k to $\mathbb{C}$ does not change their functional transcendence degree.*

Proof. Fix finite bounds on the degrees in z and in the functions. The coefficients of an identity in this finite monomial space satisfy a system of linear equations over k , obtained by Taylor coefficient comparison. Its row space is finite dimensional, so finitely many equations suffice. The complex kernel is the scalar extension of the k -kernel. Clearing rational denominators and varying the bounds proves the assertion. $\square$

For $\xi \in k^\times$, put

$$K_{\partial, \xi} = k(P_0(\xi), P_1(\xi), Q_0(\xi), E_a(\xi), E_b(\xi)).$$

It is the field of all polynomial-amplitude values on the three fixed faces and their endpoints: the one-variable reductions have only powers of z in their denominators.

Theorem 13.2 (Algebraic specialization). *For every $\xi \in \overline{\mathbb{Q}}^\times$, the numbers $M_0(\xi), M_1(\xi)$ are algebraically independent over $K_{\partial, \xi}$. Consequently*

$$K_{\partial, \xi}[I_D(r dx \wedge dy)(\xi) : r \in k[x, y]] \simeq K_{\partial, \xi}[X_0, X_1].$$

Proof. All coordinates of the boundary system and of its augmentation by M_0, M_1 are E -functions, and the combined rational system has no finite singular point other than zero. Let

$$d = \mathrm{trdeg}_{k(z)} k(z)(P_0, P_1, Q_0, E_a, E_b).$$

By Theorem 12.2 and Lemma 13.1, adjoining M_0, M_1 raises this degree to $d + 2$. The Siegel–Shidlovskii equality, in the form recalled in [2, Theorem 1.1], applied first to the boundary vector and then to the combined vector, gives degrees d and $d + 2$ at every $\xi \neq 0$. The two additional values therefore have transcendence degree two over $K_{\partial, \xi}$. The normal-form coefficients are Laurent polynomials in z , so specialization proves the second assertion. $\square$

Corollary 13.3 (Value-level twisted exactness). *For $r \in k[x, y]$ and $\xi \in k^\times$, the following are equivalent:*

1. $I_D(r dx \wedge dy)(\xi)$ is algebraic over $K_{\partial, \xi}$;
2. it belongs to $K_{\partial, \xi}$;
3. $r dx \wedge dy \in (d + \xi dq \wedge) \Omega_{k[x, y]/k}^1$.

Proof. Specialize (9.4). A nonzero linear form in two algebraically independent variables is transcendental over their base field. Thus the first condition forces $a_r(\xi) = b_r(\xi) = 0$, and specialized polynomial division makes this equivalent to the third condition. Literal Stokes gives the second condition in that case. $\square$

For clarity, let $R_\partial = k[z, P_0, P_1, Q_0, E_a, E_b]$ and $B_\xi = (R_\partial)_{(z-\xi)}$. The boundary-DVR lemma Lemma 14.1 gives residue field $K_{\partial, \xi}$. By the functional and numerical independence just proved,

$$B_\xi[M_0, M_1] \simeq B_\xi[X_0, X_1], \quad \ker(\text{ev}_\xi) = (z - \xi)B_\xi[X_0, X_1].$$

Numerical injectivity holds on the fiber $K_{\partial, \xi}[X_0, X_1]$, not on the unspecialized algebra.

Remark 13.4 (Formal period operations). The actual functional and numerical independence theorems above do not use an unpublished completeness theorem for the boundary. To identify all relations among boundary symbols with Kontsevich–Zagier operations, use the separate compact polynomial-line formal comparison of Paper III [11]. Work over the actual boundary field functionally, and over the boundary localization at elements nonzero at ξ for values. Use the compact polynomial-line formal comparison to identify that coefficient algebra with its actual realization. Then (9.4), ordinary twisted Stokes, and polynomial freeness prove that every relation in this localized fixed- D, q amplitude algebra is generated by those boundary relations and the explicit Stokes reductions. This does not assert an unlocalized cancellation theorem for an arbitrary symbol ring. There is no remaining determinant generator, no assumed isotypic normal-function compatibility, and no need to infer arithmetic constants from punctual support.

14 Local specialization without a global boundary normal form

This section is independent of the special elliptic calculation. It separates arithmetic lifting from the geometric realization of a generic relation ideal.

Lemma 14.1 (Boundary evaluation and its DVR). *Let $f_1, \dots, f_N$ be E -functions forming a closed rational first-order system over $k(z)$; include 1 if necessary. Suppose $\xi \in k^\times$ is nonsingular for that system. Put*

$$R = k[z, f_1, \dots, f_N], \quad \pi = z - \xi.$$

Then $\ker(\text{ev}_\xi : R \rightarrow \mathbb{C}) = \pi R$. The localization $B = R_{\pi R}$ is a DVR with uniformizer π , residue field $k(f_1(\xi), \dots, f_N(\xi))$, and valuation equal to analytic vanishing order at ξ .

Proof. For $P(z, f(z))$ vanishing at ξ , Beukers' theorem supplies $Q(z, X) \in k[z, X]$ with $Q(z, f(z)) = 0$ and $Q(\xi, X) = P(\xi, X)$; homogenize by 1 if needed [2, Theorems 1.2–1.3]. Thus $P - Q = \pi H$ and $P(z, f(z)) \in \pi R$. The reverse inclusion is immediate.

The ring R is a finitely generated domain, and πR is a nonzero principal prime. The principal ideal theorem makes its height one. Consequently B is a one-dimensional noetherian local domain with maximal ideal generated by π , hence a DVR. The residue field is the fraction field of $R/\pi R = \text{ev}_\xi R$. A denominator inverted in B is nonzero at ξ , so it is an analytic unit. Repeated division by π , using the kernel statement, shows that the DVR valuation is exactly the analytic order. No global normality of R has been assumed. $\square$

Theorem 14.2 (Abstract local functional and numerical comparison). *In the setting of Lemma 14.1, let $K = \text{Frac}(B)$. Let $U = (U_1, \dots, U_n)$ be actual E -functions such that (f, U) is a closed rational first-order system nonsingular at ξ . Suppose $J \subset B[X]$ is an ideal of realized identities and satisfies*

- (i) $JK[X] = \ker(K[X] \rightarrow K[U])$;
- (ii) $B[X]/J$ is B -flat.

Then $J = JK[X] \cap B[X]$, the functional realization of $B[X]/J$ is injective, and its numerical kernel is $\pi(B[X]/J)$. In particular the numerical realization is injective on

$$(B[X]/J) \otimes_B \kappa_\xi, \quad \kappa_\xi = B/\pi B.$$

If the generators of J are proved formal identities, these conclusions are formal comparison statements in that specified localized category.

Proof. Flatness over the DVR implies torsion-freeness, hence injectivity into the generic fiber. Hypothesis (i) then gives ideal contraction and functional injectivity. Let a class have numerical value zero. Multiply it by a common coefficient denominator nonzero at ξ , so that it is represented by a polynomial $P(z, f, U)$ over k . Apply Beukers lifting to the combined closed vector $(f, U, 1)$ to obtain a functional relation Q with $Q(\xi, T, X) = P(\xi, T, X)$. After substituting f , the class of Q is zero by functional injectivity, while $P - Q$ is divisible by π . The original class therefore belongs to $\pi(B[X]/J)$, since the cleared denominator is a unit of B . Conversely every such multiple specializes to zero. This proves numerical injectivity on the fiber. $\square$

This theorem does not identify an unspecified generic ideal with geometric relations. The equality in (i), the realization of its generators, and a regular local coefficient model are separate obligations. A rational change of basis with a pole at ξ cannot be used as a polynomial coordinate automorphism over B without a further lattice argument.

15 Scope, coefficient fields, and falsification controls

15.1 What is proved and what remains conditional

The construction in Theorem 2.1 is general for polynomial phases and polynomially parametrized compact chains. The single-phase geometric theorem is on the explicitly stated observable real Lefschetz locus. The actual-comparison theorem has explicit group, projective exclusion, and nonzero-quotient hypotheses. All of these are verified for (1.1). Its numerical theorem is valid for every nonzero algebraic parameter, with no extra exceptional set, because the specified closed systems have no other finite poles.

The boundary field in this concrete theorem consists only of the three fixed faces and their endpoints. Replacing it by all compact polynomial-line periods is not an asserted extension. Likewise, one fixed chain and a basis of its rank-two twisted cohomology give one actual column. Multiple chains can give multiple affine columns. A scalar cyclic presentation alone does not count isotypic multiplicities.

For arbitrary isotypic or multiblock collections, the abstract orbit result in the appendix is available once its actual group hypotheses hold. Geometric realization of the affine-linear identities and rank-one identifications is a separate comparison problem. No general numerical formal-completeness theorem for such collections is asserted here merely from the previous endpoint-complete terminology.

15.2 Four distinctions that cannot be suppressed

The parameter must be specialized. The element $z - \xi$ is nonzero as a functional germ and vanishes numerically. Thus numerical injectivity is a statement about a fiber algebra, never the unspecialized algebra containing z .

Punctual support is not arithmetic normalization. A point-supported distribution is $\sum c_m \delta_a^{(m)}$ and has Laplace transform $\sum c_m (-z)^m e^{az}$. Algebraicity of a does not imply algebraicity of c_m . For example near the minimum $(-1, 0)$ of $q = y^2 - x^3 + 3x$, writing $u = x + 1$ gives $q + 2 = 3u^2 - u^3 + y^2$. The local analytic substitution $X = u\sqrt{3 - u}$ has inverse area Jacobian $1/\sqrt{3}$ at zero. A small rational rectangle around that minimum has phase density $\mathbf{1}_{t > -2} g(t + 2)$ with $g(0) = \pi/\sqrt{3}$. Its derivative contains $(\pi/\sqrt{3})\delta_{-2}$. This coefficient is not in k . The unsplit polynomial Stokes system remains arithmetic; the counterexample concerns independently normalized pieces, not that system.

Actual boundary rings need not have the old global presentation. Let $h = (e^z - 1)/z$ and include the entire companion $(1 - e^{-z})/z$. Their ring with $z, e^{\pm z}$ is

$$k[z, h, (1 + zh)^{-1}].$$

It has the nonconstant unit $1 + zh = e^z$, which becomes 1 modulo z . A Laurent-polynomial algebra followed by a polynomial extension over $k[z]$ has no such unit. The local DVR lemma, not that global normal-form claim, is what the endpoint proof needs.

Do not clear denominators by nonunits at the endpoint. Over B with uniformizer π , the equations $\delta - 1 = 0$ and $\pi(\delta - 1) = 0$ have the same generic ideal but different local and special-fiber ideals. All coefficients of local formal generators must be normalized before specialization. In particular, ratios representing a unit must use denominators nonzero at ξ .

15.3 Stokes matrices and complex Hodge maps

For $N = E_{12} + E_{23}$, the multiplier e^N has a nonzero E_{13} entry but its logarithm N does not. Root extraction must use actual logarithms or a separately justified factorization into root subgroups. Distinct nonzero torus weights then isolate the components, and a strongly connected root graph generates $\mathfrak{sl}_n$ by elementary brackets. We do not infer the logarithmic support entrywise from a Stokes multiplier.

Similarly $\sqrt{2}\text{id}$ is a complex flat Hodge intertwiner, but no positive integer multiple becomes a rational homology map. Rational morphisms and complex linear combinations of correspondences are distinct assertions, as stated in Proposition 7.4 and Corollary 7.5.

15.4 Exact certificates and analytic audit

The repository script `scripts/check_paper_v.py` verifies the polynomial reductions, actual connection certificates, rational Gelfand–Leray identities, Fourier matrix identity, scalar Bessel equations, and independent triangle/face Taylor coefficients. The focused audit note records the branch continuation, the nonzero nodal logarithm, the Picard–Vessiot tensor and normal-subgroup arguments, and the coefficient fields in specialization. Finite algebraic checks do not constitute a kernel-checked proof of the analytic or Galois arguments. The separate certificate routine produces a checkable polynomial identity for a supplied rational-coefficient amplitude.

A Affine orbit ideals and unit-normalized flat models

This appendix retains the abstract algebraic mechanism for future multicolumn applications. It does not assign geometric realizations to relations solely because they are differential-Galois invariants.

Let K be a differential field of characteristic zero with algebraically closed constants C_0 . For each block, let

$$Y' = AY, \quad U' = AU + B_0, \quad U \in \text{Mat}_{n \times r}, \quad B_0 \in K^{n \times r}.$$

Assume $n \geq 2$, and that the reductive homogeneous group of all the blocks contains the product of their standard special-linear groups. All matrices are taken in a common no-new-constants Picard–Vessiot extension.

Proposition A.1 (Affine matrix reduction). *Constant column operations split off translation-active columns. The relation ideal is the extension of its contraction to the remaining columns; the active coordinates are polynomially free over those columns. Every remaining split block has the form $U_{\text{sp}} = A_0 + YC$, where A_0 has entries in K and C is constant. Further constant column operations discard precisely the kernel of C as affine-linear relations.*

Proof. The translation kernel is an invariant additive subgroup, hence a vector subspace in characteristic zero. On a standard isotypic block it has form $C_0^n \otimes R$, since the standard representation is simple with scalar endomorphisms. Choose columns adapted to R . Translations add Yv independently to each active column; their orbit is dense in its affine coordinate space, so the ideal is extended from the other coordinates. On the quotient columns the affine cocycle descends to the reductive homogeneous group. Complete reducibility splits the extension of the trivial representation by that module, so the cocycle is a coboundary. The shifted solution is Galois-fixed and belongs to K . Finally perform ordinary constant column reduction on C . $\square$

For a constant $n \times s$ matrix C of column rank $s < n$, the SL_n -orbit is the full rank- s locus, hence dense: any GL_n map between two such matrices can have its determinant corrected on a complementary vector. If $s = n$, its orbit is exactly $\det X = \det C$. Thus only square full-rank split blocks remain nonlinear. For these write $\Delta_\lambda = \det(U_\lambda - A_\lambda)$. Let $T_{\det} \subset \mathbb{G}_m^s$ be the determinant image of the joint group and

$$L = \{a \in \mathbb{Z}^s : \chi^a|_{T_{\det}} = 1\}, \quad \eta_a = \Delta^{a^+} / \Delta^{a^-} \in K^\times.$$

The subgroup can be disconnected, so L need not be saturated.

Theorem A.2 (Abstract complete affine orbit ideal). *Under the stated product-group hypotheses, all relations are generated by the affine-linear reductions and the full lattice ideal*

$$\Delta^{a^+} - \eta_a \Delta^{a^-}, \quad a \in L.$$

Proof. After the linear and translation reductions, independent special-linear factors remove every invariant except the square-block determinants. The remaining orbit is a translate of the determinant subgroup in a torus. Its Laurent ideal is the character ideal; contraction to the ordinary polynomial ring is the partial-character lattice ideal with all $a \in L$ [8]. Pullback under the determinant maps and adjoining the free coordinates gives the full orbit-closure ideal. This pullback is reduced: determinant morphisms are flat with geometrically integral fibers, and the characteristic-zero translated subgroup is reduced. The same facts show that the preimage of its dense torus is dense. The Picard–Vessiot torsor identifies the descended orbit ideal with the algebraic relation ideal of the actual solution coordinates. $\square$

Now work over a DVR B at a nonsingular parameter. Assume the matrix frames and coordinate transformations are regular and invertible over B . Use an actual fundamental matrix holomorphic and invertible at the point. A split shift $U - YC$ belonging to $K = \text{Frac}(B)$ is then holomorphic, hence in B when the DVR valuation is analytic order. Square-block determinants have nonzero value, so $\eta_a \in B^\times$. This is an assertion about such a specified regular model, not about an arbitrary meromorphic frame.

Lemma A.3 (Unit-valued lattice quotients). *For a character $\rho : L \rightarrow B^\times$, the algebra*

$$D_\rho = B[\delta_1, \dots, \delta_s] / (\delta^{a^+} - \rho(a)\delta^{a^-} : a \in L)$$

is free as a B -module.

Proof. Partition $\mathbb{N}^s$ by differences in L . Choose u_c in each class c . Map δ^u to $\rho(u - u_c)e_c$ in the free module with basis e_c . The character law makes every defining binomial vanish. Conversely any two monomials in the same class differ, up to the prescribed unit, by a monomial multiple of the defining binomial for their difference. Thus the kernel is exactly the lattice ideal, and the e_c give a free basis. $\square$

Lemma A.4 (Free determinant morphism). *The ring $B[x_{ij}]$ is free over $B[\delta]$ under $\delta \mapsto \det(x_{ij})$.*

Proof. Present it as $B[\delta, x_{ij}] / (\det X - \delta)$. A lexicographic order in the matrix variables gives a determinant monomial leading coefficient 1. Division by this single monic polynomial has unique remainder in monomials not divisible by its leading monomial. These form a free $B[\delta]$ -basis. This proof is valid over an arbitrary coefficient ring. $\square$

Corollary A.5 (Flat regular orbit models). *For regular affine shifts and unit-valued determinant characters as above, the orbit quotient over B is B -flat and its ideal is the contraction of the generic orbit ideal.*

Proof. Remove coordinate affine-linear relations and translate by the shifts in B . The quotient is obtained from D_ρ by base change under the product of the free determinant morphisms, followed by polynomial extension in the other coordinates. It is therefore B -flat. Torsion-freeness gives contraction from the generic fiber. $\square$

Choose representatives $\eta_a = b_a/c_a$ only with b_a, c_a units at the chosen point. Then the cleared determinant equations generate the same local ideal; arbitrary multiplication by a vanishing denominator is not allowed. A finite set must generate the *full polynomial lattice ideal*, not merely the additive lattice: noetherianity supplies such a set. Geometric or formal realization of the affine shifts and of every required determinant-line identification remains an extra obligation in applying Theorem 14.2.

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